

DVWA Ubuntu Server Hardening

Final Lab Report

Item	Details
Project	DVWA Ubuntu Server Hardening
Server Hostname	UbunServDVWA-TeamA
Server IP Address	10.10.10.223/16
Platform	Proxmox Virtual Environment
Lab Type	Cybersecurity / SOC Capstone Lab
Report Purpose	Document Linux hardening, monitoring, validation, and evidence collection activities

Disclaimer

This report documents work performed in an authorized lab environment for educational and professional development purposes. DVWA is intentionally vulnerable and was used only for controlled cybersecurity training and defensive hardening practice.

1. Executive Summary

This project focused on hardening and documenting an Ubuntu-based DVWA server hosted in a Proxmox virtual lab. The server, named UbunServDVWA-TeamA, was used as an intentionally vulnerable web application target for authorized cybersecurity training. The goal was not to remove all vulnerability from DVWA, but to improve the security of the underlying Ubuntu server while preserving the ability to conduct controlled testing.

The project included system baseline review, package updates, account review, root account lockout, SSH hardening, firewall configuration, Fail2Ban protection, AIDE file integrity monitoring, Lynis auditing, Chrony time validation, unnecessary service reduction, sysctl network hardening, log review, network testing, and evidence collection.

The final result was a better documented and more defensible Ubuntu lab server with repeatable commands, a checklist, evidence outputs, and a GitHub-ready portfolio structure.

2. Lab Environment

Component	Description
Virtualization Platform	Proxmox Virtual Environment
Server	Ubuntu Server running DVWA
Hostname	UbunServDVWA-TeamA
IP Address	10.10.10.223/16
Testing Machine	Kali Linux VM using DHCP
Client Machine	Windows VM using DHCP
Primary Services	Apache, MySQL, SSH, DVWA
Security Tools	UFW, Fail2Ban, AIDE, Lynis, Chrony, tcpdump, Wireshark, Nmap

3. Project Objectives

- Establish a baseline of the Ubuntu DVWA server.
- Review user accounts, services, open ports, and network configuration.
- Harden SSH to reduce unauthorized remote access risk.
- Configure firewall rules using UFW.
- Deploy Fail2Ban to detect and block repeated SSH authentication failures.
- Deploy AIDE for file integrity monitoring.
- Run Lynis audits to identify hardening opportunities.
- Validate time synchronization for accurate logging and event correlation.
- Reduce unnecessary services to lower attack surface.
- Apply sysctl network hardening settings.
- Collect evidence and create portfolio-ready documentation.

4. Work Performed

4.1 System Baseline Review

- Reviewed hostname, operating system, kernel version, uptime, logged-in users, IP configuration, routing, DNS resolver status, listening ports, running services, and failed services.
- Commands included hostnamectl, ip a, ip route, ss -tulnp, lsof -i, systemctl, and journalctl.

4.2 Package Updates and Maintenance

- Updated package lists and upgraded installed packages using apt.
- Removed unnecessary packages and cleaned package cache to reduce maintenance overhead.

4.3 User and Account Review

- Reviewed local users, login shells, groups, sudo membership, and password aging details.
- Locked the root account to reduce the risk of direct privileged account misuse.

4.4 SSH Hardening

- Backed up SSH configuration before changes.
- Reviewed or applied settings for root login restrictions, authentication controls, maximum login attempts, idle timeout behavior, TCP forwarding, X11 forwarding, and login banner.
- Validated syntax with sshd -t before restarting SSH.

4.5 Firewall Configuration

- Configured UFW with deny incoming and allow outgoing default policies.
- Allowed SSH access and reviewed HTTP access based on DVWA lab requirements.
- Saved firewall status for evidence.

4.6 Fail2Ban SSH Protection

- Installed and enabled Fail2Ban.
- Configured the SSH jail to detect repeated authentication failures.
- Reviewed jail status and Fail2Ban logs.

4.7 AIDE File Integrity Monitoring

- Installed AIDE and initialized the database.
- Copied the new database into the active database location.
- Ran AIDE checks to validate file integrity monitoring capability.

4.8 Lynis Security Audit

- Installed Lynis and ran a system audit.
- Reviewed warnings, suggestions, and hardening index results to guide additional improvements.

4.9 Chrony Time Synchronization

- Installed and validated Chrony.
- Reviewed tracking, sources, and timedatectl output.
- Confirmed time synchronization for reliable log review.

4.10 Service Reduction

- Reviewed services such as Avahi, CUPS, Whoopsie, and LightDM.
- Disabled services that were not required for the server role where appropriate.

4.11 Sysctl Network Hardening

- Backed up sysctl configuration.
- Reviewed settings for reverse path filtering, redirect handling, bogus ICMP handling, and TCP SYN cookies.
- Applied and validated configuration changes.

4.12 Log Review and Monitoring

- Reviewed system logs, authentication logs, failed SSH attempts, accepted SSH logins, sudo activity, and Fail2Ban logs.
- Used logs to support SOC-style monitoring and security event review.

4.13 Network Testing and Traffic Capture

- Used ping, ip neigh, Nmap, tcpdump, and Wireshark to validate connectivity and collect traffic evidence.
- Testing was performed only in the authorized lab environment.

5. Security Control Summary

Control Area	Action Taken	Security Value
System Baseline	Reviewed host, network, service, and log state.	Establishes known starting point.
Account Security	Reviewed users and locked root account.	Reduces privileged account misuse risk.
SSH Hardening	Restricted SSH behavior and validated config.	Reduces remote access attack surface.
Firewall	Configured UFW default policies and service rules.	Limits inbound exposure.
Fail2Ban	Enabled SSH jail and reviewed logs.	Reduces brute-force risk.
AIDE	Initialized file integrity database.	Supports unauthorized change detection.
Lynis	Ran system audit and reviewed findings.	Identifies hardening gaps.
Chrony	Validated time synchronization.	Improves log accuracy.
Service Reduction	Reviewed and disabled unnecessary services.	Reduces attack surface.
Sysctl	Applied network hardening settings.	Improves network-layer resilience.

6. Issues Encountered and Resolutions

Issue	Resolution
AIDE initialization required additional steps.	Initialized the AIDE database and copied aide.db.new to the active database path.
Fail2Ban required proper local jail configuration.	Created and reviewed jail.local, restarted Fail2Ban, and validated sshd jail status.
SSH hardening could risk administrative lockout.	Backed up configuration and tested syntax with sshd -t before restart.
DVWA requires web access for testing.	Balanced HTTP firewall rules with the current lab objective.
Time synchronization needed validation.	Configured and reviewed Chrony tracking and timedatectl status.

7. Evidence Collected

The following evidence files are recommended for the project repository. They support auditability, repeatability, and professional documentation.

- hostnamectl.txt
- ip-addresses.txt
- ip-route.txt
- listening-ports.txt
- ufw-status.txt
- fail2ban-status.txt
- fail2ban-sshd-status.txt
- chrony-tracking.txt
- running-services.txt
- failed-services.txt
- Screenshots of UFW, Fail2Ban, AIDE, Lynis, Chrony, Apache status, and listening ports.

8. Final Outcome

The DVWA Ubuntu server was successfully reviewed, hardened, and documented. The completed project demonstrates practical Linux security administration and SOC-style defensive work. The server retained its purpose as a controlled vulnerable web application for training while gaining stronger administrative controls, monitoring visibility, and supporting documentation.

Key deliverables include README.md, commands.txt, hardening-checklist.md, final-lab-report.pdf, evidence outputs, and screenshots.

9. Reflection

This project reinforced the importance of balancing security with operational requirements. Because DVWA is intentionally vulnerable, the goal was not to eliminate all exposure, but to secure the surrounding Ubuntu server, protect administrative access, improve logging, and document every meaningful action. The project also showed the value of repeatable commands, validation checks, and evidence collection for cybersecurity reporting.

10. Recommended GitHub Folder Structure

```
DVWA-Ubuntu-Server-Hardening/  
  README.md  
  commands.txt  
  hardening-checklist.md  
  final-lab-report.pdf  
  evidence/  
    hostnamectl.txt  
    ip-addresses.txt  
    listening-ports.txt  
    ufw-status.txt  
    fail2ban-status.txt  
    chrony-tracking.txt  
  screenshots/  
    ufw-status.png  
    fail2ban-sshd-status.png
```

aide-check.png
lynis-summary.png
chrony-tracking.png